

GOVERNMENT OF INDIA	DEEPPFAKE FORENSIC SUMMARY
BharatShield National Deepfake Detection and Response Platform	Reference No.: DFS-0c1f41ce
RESTRICTED – LAW ENFORCEMENT USE ONLY	

Deepfake Forensic Summary Report

BharatShield Multi-Modal Forensic Detection Benchmarks

1. CASE DETAILS

Packet Reference	DFS-0c1f41ce
Case Identifier ID	0c1f41ce-15a0-47a9-9e62-830a24103df6
Generated Date	04-06-2026 21:00:08 IST
Subject Name Depicted	Shri Demo Politician
Media File SHA-256	b94d27b9934d3e08a52e52d7da7dae55cfe5f806fb7c93f00039b7c69b6c4220

RESOLVED SUBJECT BIOMETRICS & IDENTITY PROFILE

Match Status	RESOLVED / BIOMETRIC MATCH DETECTED
Resolved Subject Name	Shri Demo Politician
Aadhaar Number (Masked)	XXXX-XXXX-7890
Gender	Female
Electoral Role	active_candidate
Constituency	Demo Lok Sabha Constituency
Party Affiliation	Demo National Party
Active Candidacy (MCC)	Yes
Face Cosine Similarity	100.0000%
Voice Cosine Similarity	100.0000%
Fused Confidence Score	100.0000%
Verification Source	hybrid

2. DETAILED FINDINGS

Summary: Media classified as likely synthetically manipulated based on 12 forensic indicator(s), including 4 high-severity finding(s).

Category	Severity	Finding Details	Metric Reference
Audio-Video Sync	CRITICAL	Audio–video lip movement is misaligned, consistent with synthetic dubbing or post-hoc voice replacement rather than naturally captured speech.	lip_sync_alignment_error_ms: 135.0 ms
Facial Manipulation	HIGH	Spatial facial inconsistencies detected across frames suggest GAN or deepfake-based facial manipulation.	spatial_cnn_manipulation_probability: 92.00%
3D Face Geometry	MEDIUM	3D face mesh landmarks are unstable across the temporal sequence, indicating inconsistent facial geometry typical of synthetic faces.	face_mesh_landmark_variance: 0.0750
Synthetic Voice	HIGH	The voice track exhibits characteristics consistent with text-to-speech or neural voice cloning rather than a live recording.	tts_synthetic_probability: 88.00%
Voice Authenticity	MEDIUM	Pitch contour and spectrogram features do not match natural speech patterns for the apparent speaker, suggesting audio synthesis or splicing.	spectrogram_pitch_mismatch_ratio: 39.00%
Anti-Spoofing	HIGH	Anti-spoofing neural network assigns low confidence to bona fide speech, indicating the audio is likely non-genuine or heavily processed.	anti_spoofing_nn_confidence: 11.00%
Lighting Consistency	MEDIUM	Lighting on the face and background shifts unnaturally between frames, suggesting compositing or generative synthesis rather than a single capture.	illumination_inconsistency_score: 71.00%
Shadow Geometry	MEDIUM	Shadow direction and softness are inconsistent with the scene light source, a common artifact in deepfake and face-swap videos.	shadow_geometry_inconsistency_score: 66.00%
Color Grading	MEDIUM	Skin tone and color grading do not remain consistent across frames; color boundaries and shadows appear imperfect relative to natural video.	color_grading_anomaly_score: 59.00%
Temporal Stability	MEDIUM	High-frequency flicker and temporal instability detected at facial boundaries, consistent with frame-wise generative manipulation.	temporal_flicker_score: 55.00%
Facial Boundaries	MEDIUM	Visible artifacts at the jawline, hairline, or face perimeter suggest blending of a synthetic face onto an original body.	facial_boundary_artifact_score: 62.00%
Analyst Observation	MEDIUM	Dummy run: lip-sync mismatch; inconsistent lighting and shadows on face.	—

CDAC Principal Scientist Expert Witness Representative AI Forensics Division, CDAC	Signature: _____ Date: 04 June 2026
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